

Cross-Family Solver Consensus Outperforms LLM Judges for Logic Translation Evaluation

Abstract

Evaluating whether a first-order logic (FOL) formula faithfully captures a natural-language sentence is difficult without a reference formula. We propose cross-family solver consensus: translating each sentence with several independent LLM families, checking pairwise equivalence via an SMT solver, and scoring each candidate by its agreement with the pool. On a held-out set of 2,686 real NL-to-FOL translations from nine model families, consensus achieves stratified AUROC 0.741, exceeding the best cheap LLM judge by +0.099 (95% CI [0.049, 0.146]). The advantage grows on long, conditioned sentences and shrinks on short ones. Three peer families suffice for nearly all discriminative power at a marginal cost below \$0.001 per candidate. The metric’s main limitation is vocabulary dependence: candidates using different predicate names than their peers are flagged regardless of correctness, and no variant we tested resolved this without losing discriminative power.

1 Introduction

Translating natural language into first-order logic (autoformalization) is a prerequisite for neurosymbolic reasoning pipelines [Wu et al., 2022, Pan et al., 2023], yet the evaluation problem remains open. A correct FOL formula for a sentence is not unique: different predicate names, different decompositions of the same concept, and logically equivalent restatements can all be faithful. Gold FOL annotations are scarce and unreliable. Recent audits found that 36–42% of annotations in the FOLIO [Han et al., 2022] and MALLS benchmarks contain errors [Brunello et al., 2026]. Existing metrics either require a gold formula (exact match, prover-checked equivalence) or verify only that the output parses [Ganguly et al., 2025, Ryu et al., 2024].

The gap is a metric that predicts whether a formula is faithful to the text, given no reference formula, no ontology, and no controlled vocabulary. An LLM judge can fill this role, but judges degrade on long, complex sentences, are sensitive to meaning-preserving rewrites of the formula [Amrollahi et al., 2026], and cannot verify logical equivalence. A trained verifier [Singh et al., 2026] needs supervised labels from the same gold annotations whose quality is in question.

We study cross-family solver consensus as a gold-free faithfulness metric. If multiple independent LLM families translate the same sentence and a candidate’s formula is z3-equivalent to what most peers produce, the candidate is likely correct; when a formula disagrees with the consensus, it is likely wrong. The score is the fraction of family-disjoint peers whose translations are not z3-equivalent to the candidate, after vocabulary alignment.

This paper makes three contributions:

1. We show that cross-family solver consensus achieves stratified AUROC 0.741 on a held-out development set of 2,686 real translations (nine families), exceeding the best cheap LLM judge (Gemini 2.5 Flash Lite, disguised) by +0.099 [0.049, 0.146] and adding +0.039 [0.021, 0.057]

over the full baseline stack of 28 cheap metrics. On long rule sentences with trusted references, consensus reaches 0.954 versus the judge’s 0.587.

2. We construct a meta-evaluation suite of 8,507 real candidate formalizations from nine LLM families on 700 sentences, with labels from z3 equivalence to audited gold and a three-member adjudication panel, plus a typed perturbation suite of 4,234 controlled mutations and 868 meaning-preserving controls.
3. We characterize the metric’s failure modes. Consensus is fundamentally vocabulary-dependent: any predicate renaming breaks z3 equivalence, producing false-alarm rates above 0.60. Error endorsement, where a majority of peers reproduce the same mistake, sets a recall ceiling of approximately 62%. Three peer families capture 97% of full-pool performance at a cost of \$0.002 per sentence.

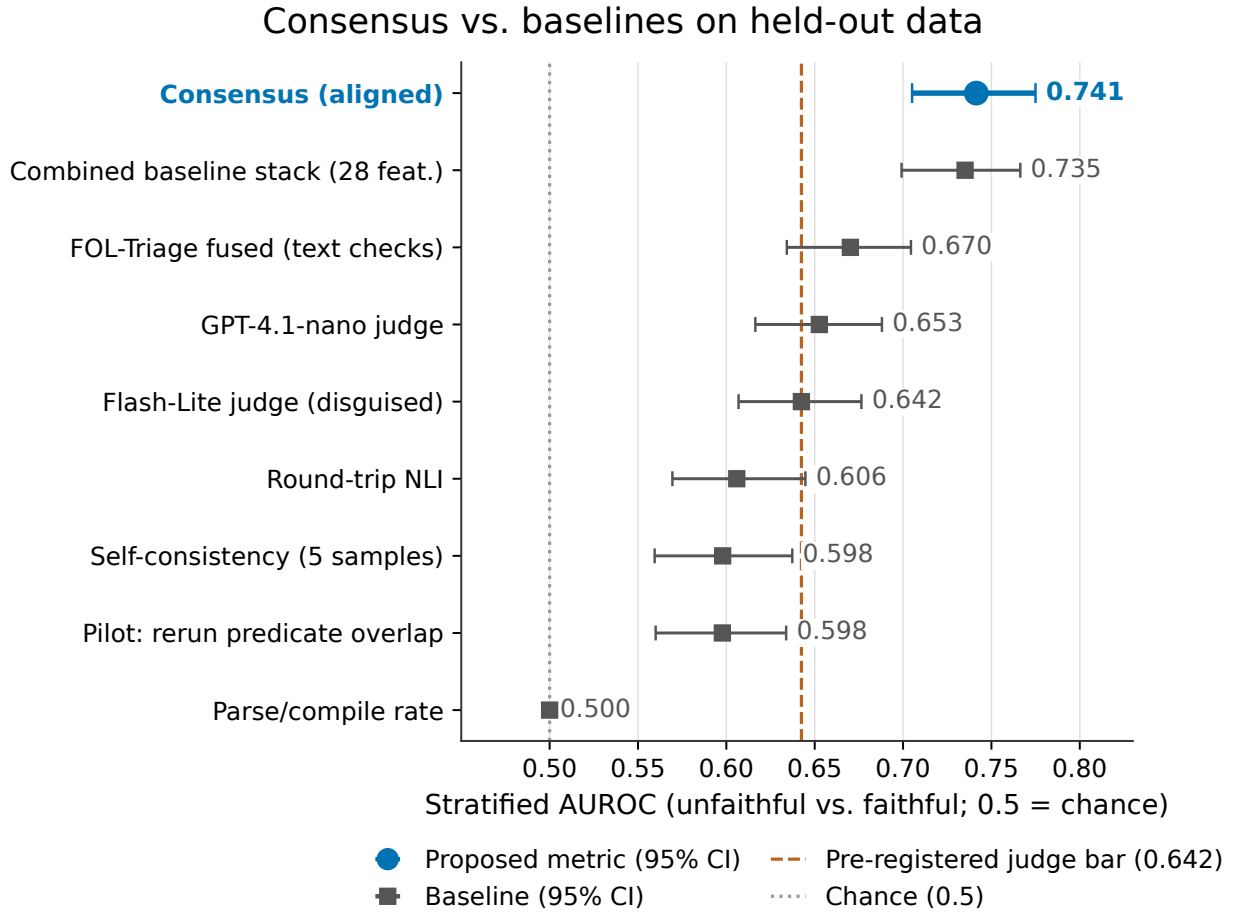


Figure 1: Stratified AUROC of cross-family solver consensus and baseline faithfulness metrics on the held-out development set (2,686 candidate formalizations, 1,822 unfaithful / 864 faithful, 292 sentences). Whiskers are 95% sentence-clustered bootstrap CIs ($B = 2000$). The dashed line marks the pre-registered judge bar (Flash-Lite judge, disguised view, 0.642) and the dotted line marks chance (0.5). Consensus scores highest (0.741) and exceeds the judge bar by +0.099 [0.049, 0.146]. Adding it to the combined baseline stack raises AUROC by +0.039 [0.021, 0.057], so it carries signal the baselines lack.

2 Related Work

Gold-free evaluation of formal translations. Several recent systems evaluate autoformalization output without requiring a reference formula. Amrollahi et al. [2026] propose roundtrip verification: verbalizing the formula back to natural language and checking NLI entailment with the original sentence. Rules that fail show $1.4\text{--}2.5\times$ more NLI drift, but the method does not produce a per-candidate score. GenV [Singh et al., 2026] distills a z3-equivalence oracle into a reference-free reward model for NL-to-FOL, achieving AUROC 0.961 on z3-reference labels. When the label target switches from solver equivalence to panel-judged intent, the LLM judge beats GenV (0.778 vs. 0.679), illustrating the sensitivity of metric rankings to the label protocol. FormalAlign [Lu et al., 2024] trains an alignment scorer for Lean autoformalization. GoFU [Ganguly et al., 2025] uses PCFG-based uncertainty over SMT-LIB programs from a single model’s samples, enabling selective verification that cuts error rates by 14–100%. These methods either rely on a single model’s distribution or require supervised training on gold labels.

Cross-model consensus. The idea that agreement among independently generated solutions signals correctness has been studied in software engineering as N-version programming [Eckhardt and Lee, 1985, Knight and Leveson, 1986] and in LLM reasoning as cross-model consistency [Du et al., 2023, Liu, 2026, Zhang et al., 2023]. SAC³ [Zhang et al., 2023] uses cross-model cross-question consistency for hallucination detection in QA. Liu [2026] formalize a generative model of cross-family consensus for reasoning, showing that wrong answers scatter while the correct answer accumulates agreement; they recommend 3–4 cross-family models. In hardware verification, NoTB [Alvanaki et al., 2026] clusters RTL translations from four LLM families by formal equivalence, achieving 94.7% precision at 27% coverage on 78 specifications. For NL-to-temporal-logic, Wang et al. [2026] compare single-model self-consistency against judges and back-translation by difficulty tier. ARc [Bayless et al., 2025] redundantly translates natural language to SMT-LIB with several LLMs and scores each translation by the share of others that entail it, the same functional form as our consensus score, but over a fixed schema without vocabulary alignment.

Our work differs from these predecessors in three ways. First, we evaluate cross-family consensus as a per-candidate faithfulness score for NL-to-FOL, where vocabulary divergence across families is the central obstacle, not a controlled variable. Second, we conduct a head-to-head meta-evaluation against LLM judges, round-trip methods, self-consistency, and structural baselines on adjudicated labels, with pre-registered hypotheses. Third, we decompose consensus performance into error endorsement and correct-candidate divergence, connecting it to the coincident-failure theory of Eckhardt and Lee [1985] and Knight and Leveson [1986].

Self-consistency for formal languages. Single-model self-consistency (sampling K translations from the same model and checking their agreement) is a natural baseline. Li et al. [2024] select among K temperature samples by symbolic equivalence for mathematical autoformalization. Chen et al. [2023] propose universal self-consistency. CLOVER [Ryu et al., 2024] notes that incorrect FOL formulas are less likely to be equivalent, but also observes consistent last-step mistakes. On our data, five-sample self-consistency from a single model (gpt-4.1-nano) achieves AUROC 0.725 [0.669, 0.781], significantly below cross-family consensus at 0.866 [0.821, 0.906], a gap of +0.140 [0.085, 0.197].

3 Method

3.1 Problem Setting

Given a natural-language sentence s and a candidate FOL formula c produced by any system, we want a score $f(s, c) \in [0, 1]$ that predicts whether c faithfully captures the meaning of s , without access to a reference formula or ontology. Higher scores indicate greater likelihood of error.

3.2 Peer Pool Generation

For each sentence s , we collect translations from K independent LLM families. Each family receives s under a fixed few-shot prompt with temperature 0, producing one canonical translation per family. In our experiments, $K = 9$ families span Llama-3.3-70B, Qwen3-235B, DeepSeek-V3.1, Mistral-Small-3.2, Gemini-2.5-Flash-Lite, GPT-4.1-Mini, GPT-3.5-Turbo, GPT-4, and text-davinci-003. The first six are fresh peer families; the last three are the original Logic-LM [Pan et al., 2023] systems whose outputs form part of the evaluation data.

3.3 Pairwise Equivalence with Vocabulary Alignment

Two FOL formulas with identical predicate names can be checked for logical equivalence by an SMT solver. We use z3 [de Moura and Bjørner, 2008] to determine whether $c_1 \Leftrightarrow c_2$ is valid. When formulas use different predicate names, which is the common case across model families, we apply a vocabulary alignment step: a greedy name-similarity mapping based on Levenshtein distance between predicate and constant names, constrained by arity consistency. After alignment, z3 checks equivalence under the mapped names. Each pairwise check takes a median of 5 ms.

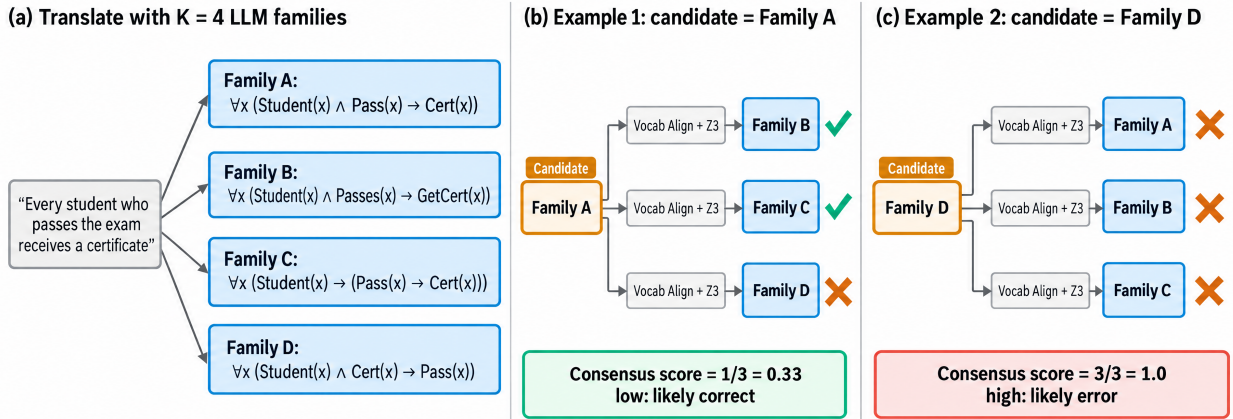


Figure 2: How consensus scoring works. (a) A sentence is translated by $K = 4$ LLM families; families A–C produce logically equivalent formulas (up to vocabulary), while family D swaps the condition and conclusion. (b) Candidate from family A: vocabulary-aligned z3 equivalence with B, C succeeds (✓) and fails with D (✗), giving a consensus score of $1/3 = 0.33$ (low, likely correct). (c) Candidate from family D: equivalence fails with all three peers, giving a score of $3/3 = 1.0$ (high, likely erroneous).

3.4 Consensus Score

For a candidate c produced by system i on sentence s , we define the consensus score as:

$$\text{c_score}(c) = 1 - \frac{|\{j \neq i : \text{eqmv}(c, p_j) = \text{true}\}|}{|\{j \neq i : p_j \text{ is parseable}\}|} \quad (1)$$

where p_j is the translation from family j and eqmv denotes z3 equivalence modulo vocabulary alignment. A candidate that agrees with all peers scores 0 (likely correct); one that agrees with none scores 1 (likely erroneous). The score is graded rather than binary: partial agreement produces intermediate values.

3.5 Decomposition: Endorsement and Divergence

The consensus score’s AUROC can be decomposed into two error terms at any binary decision threshold. Let e be the error endorsement rate: the fraction of erroneous candidates that a majority of peers endorse (agree with). Let d be the correct-candidate divergence rate: the fraction of correct candidates that a majority of peers reject (disagree with). At the majority threshold:

$$\text{AUROC}_{\text{binary}} = 1 - \frac{e + d}{2} \quad (2)$$

This decomposition separates two distinct failure modes. Error endorsement occurs when multiple families reproduce the same mistake, which is more common for structural errors (57%) than polarity errors (5%). Correct-candidate divergence occurs when a correct formula uses different vocabulary than its peers, which rises with sentence length. On our data, $e = 0.124$ and $d = 0.537$ for 9-family majority voting.

3.6 Cost and Efficiency

The full consensus score requires (a) generating peer translations and (b) running z3 pairwise checks. Generation dominates the cost: \$0.0015 per sentence for six API peers. The z3 checks are free (CPU-only, median 5 ms per pair). For a candidate already embedded in a multi-system pipeline, the marginal cost is only the z3 computation: under \$0.001 per candidate. Three families suffice for 97% of discriminative power (AUROC 0.785 vs. 0.784 for the full pool), and the optimal 3-family subset (DeepSeek + Microsoft + OpenAI) achieves OOF AUROC 0.785 at approximately \$0.002 per sentence.

4 Experimental Setup

4.1 Meta-Evaluation Datasets

Development set (E). 8,507 candidate formalizations from 10 LLM slots (9 families plus `csg2lambda`) on 700 sentences drawn from FOLIO [Han et al., 2022] training premises (CTRL stratum, 150 sentences) and long/conditional/exception strata from training data (L25: 300 sentences with ≥ 25 words and ≥ 3 conditions; L20: 150 sentences with ≥ 20 words; EXC: 100 exception sentences). Each candidate is labelled by z3 equivalence to audited gold references (tier A), with contested cases adjudicated by a three-member LLM panel (tier B). The resulting labels are 864 CORRECT, 1,822 ERROR, 1,086 UNPARSEABLE (scored as maximum error), and the remainder UNRESOLVED or CONTESTED. The primary population is R_AB (tiers A and B combined): 2,686 rows on 292 sentences.

Perturbation suite (PERTURB). 4,234 typed mutations of 300 base formulas, covering operators NEG, REV, QUANT, RESTR, CONN, MOVE, DROP, ADD, SWAP, BIND, and MEANING_RENAME, each verified as z3-non-equivalent to all accepted readings. Plus 868 meaning-preserving controls (RENAME_SYN, RENAME_NONCE, REORDER, CONTRAPOSITIVE, DEMORGAN), each z3-equivalent to the base. Used for controlled sensitivity and invariance testing.

R.COMP (long rule sentences). 221 templated sentences with ≥ 25 words, ≥ 3 conditions, and one unless/except/provided-that/only-if clause, each with trusted weak and strong readings from the template. In the SIG (signature-block) condition, generators receive the template’s symbol list, and labels are pure z3 equivalence. Translated by 10 few-shot slots (9 families) producing 1,904 testable rows.

4.2 Baselines

We compare consensus against five baseline families, all evaluated on the same held-out items:

1. **LLM judges.** Gemini 2.5 Flash Lite (rubric-based JSON scoring, 0–100), GPT-4.1-Nano ($P(\text{YES})$), and Gemini 3.1 Pro (frontier), each in original and nonce-disguised views. Disguise replaces content words in the sentence with nonsense tokens to control for memorization.
2. **Round-trip verification.** Verbalize the formula to natural language via an LLM, then measure entailment against the original sentence using DeBERTa-based NLI. The minimum of forward and backward entailment scores serves as the round-trip NLI metric.
3. **Self-consistency (SC-5).** Sample 5 translations from GPT-4.1-Nano at temperature 0.7, check pairwise z3 equivalence, and score by the agreement fraction.
4. **Text-based checks (FOL-Triage).** A three-layer cascade: L1 (formula lint: deterministic smell rules), L2 (bag-of-words coverage between text and formula predicates), and L3 (structured questionnaire: a small LLM reads the text and z3 reads the formula; their answers are compared). Fused by logistic regression fitted on a separate track.
5. **Structural pilot metrics.** Joint satisfiability conflicts, arity inconsistency, predicate-shape agreement, dangling predicates, undeclared predicates, and cross-rerun predicate-set Jaccard.
6. **S4 stack.** A cross-fitted logistic combination of all 28 cheap baseline features (judges, round-trip, self-consistency, structural metrics).

4.3 Evaluation Protocol

All comparisons use stratified AUROC as the primary statistic, computed with sentence-clustered bootstrap ($B = 2,000$) to account for within-sentence dependence. Stratification is by source stratum (L25, L20, EXC, CTRL) to remove the composition confound between sentence pools. The consensus score and all baselines were pre-registered and sealed before any labels were read (prereg sha256 c2a6cf84). The disguised cheap judge (Gemini 2.5 Flash Lite) is the pre-registered bar. Contamination is controlled by nonce-disguised scoring and difference-in-differences tests.

5 Results

5.1 Main Result: Consensus vs. Baselines on Development Set E

Table 1 presents the head-to-head comparison on the primary population (R_AB, 2,686 rows, 1,822 ERROR / 864 CORRECT, 292 sentences).

Table 1: Stratified AUROC on the primary population (R_AB). Δ is the margin over the pre-registered judge bar (Flash-Lite disguised).

Metric	Strat. AUROC	95% CI	Δ vs. judge	95% CI
Consensus (c_score_align)	0.741	[0.693, 0.787]	+0.099	[0.049, 0.146]
S4 stack (28 features)	0.700	[0.658, 0.739]	+0.058	—
GPT-4.1-Nano judge (orig)	0.653	[0.607, 0.698]	—	—
Flash-Lite judge (disguised)	0.642	[0.593, 0.690]	—	—
FOL-Triage fused	0.614	[0.567, 0.660]	—	—
Self-consistency (SC-5)	0.592	[0.548, 0.637]	—	—
Round-trip NLI	0.564	[0.522, 0.606]	—	—
Pilot structural metrics	0.500–0.523	—	—	—

The consensus score exceeds the best cheap judge by a stratified margin of +0.099 [0.049, 0.146] (all 2,000 bootstrap samples positive). It also adds incrementally over the S4 stack of all 28 baseline features: nested [S4 + consensus] minus [S4] is +0.039 [0.021, 0.057] (permutation-null 95th percentile: 0.009). The pilot structural metrics (joint conflicts, arity inconsistency, predicate-shape agreement, dangling predicates) are null on this data, with AUROCs between 0.500 and 0.523.

The pooled (unstratified) AUROCs are higher for all metrics: consensus 0.782, judge 0.696, because the long-sentence strata where consensus dominates receive more weight. We report stratified numbers throughout to avoid this composition effect.

5.2 Consensus on Long Rule Sentences

On R_COMP SIG (1,904 rows from 221 templated sentences with ≥ 25 words, ≥ 3 conditions, and trusted references), consensus reaches within-template AUROC 0.954 [0.939, 0.968] versus the disguised Flash-Lite judge at 0.587 [0.551, 0.622], a gap of +0.367 [0.328, 0.404]. This is the condition where consensus is strongest: long sentences with many conditions, where the vocabulary is constrained by the template’s signature block and errors are unlikely to be endorsed by the majority.

The advantage concentrates in the long, conditioned strata. On the EXC (exception) stratum of E, the gap is +0.197; on L20+EXC with tier-A labels, consensus reaches AUROC 0.860 versus the judge’s 0.554. On the short FOLIO CTRL stratum, the advantage is statistically insignificant (−0.069 [−0.23, 0.10]).

5.3 Comparison with Frontier Judge

On a 284-row subsample where the frontier judge (Gemini 3.1 Pro) was scored, the consensus-to-frontier frame ratio is 0.957 [0.887, 1.034], meaning consensus achieves comparable discriminative power. The frontier judge costs \$0.00365 per item; consensus costs \$0.000121 per item (full) or \$0.00000025 per item (marginal, z3 only), a factor of 30× cheaper per candidate. On R_COMP SIG, consensus (0.956) slightly exceeds the frontier judge (0.932) on a 60-row subsample; the nested [frontier + consensus] over [frontier] gain is +0.088 [0.019, 0.172].

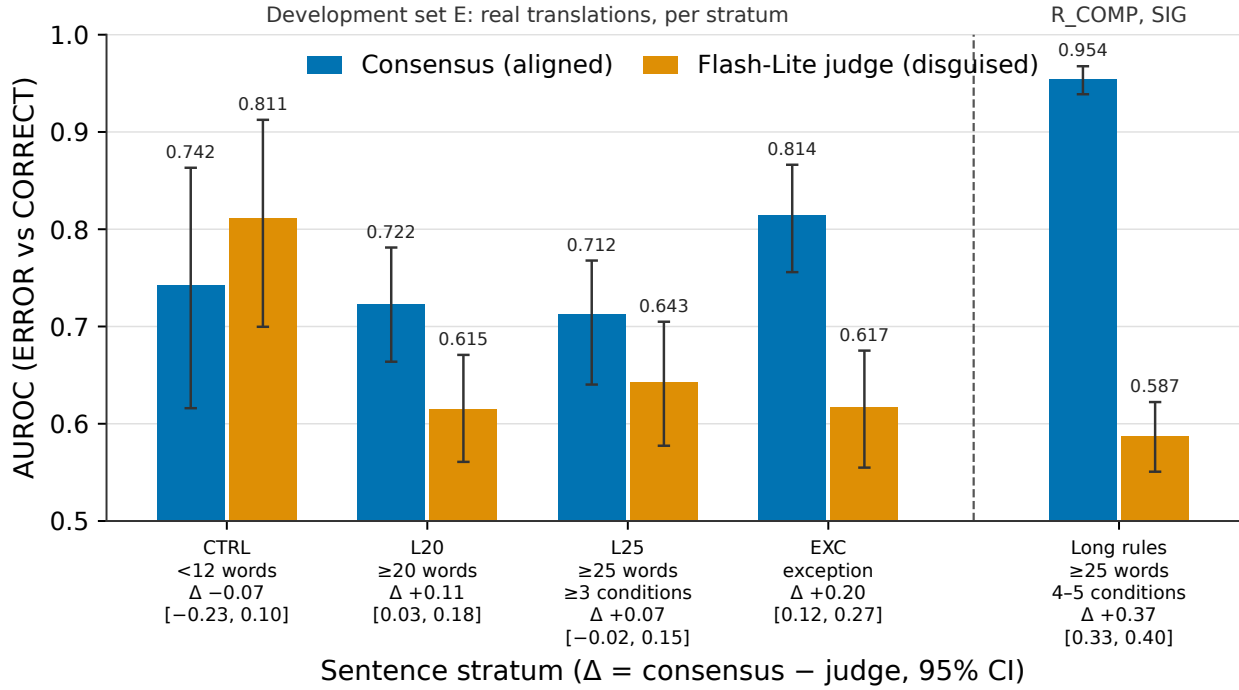


Figure 3: AUROC for consensus (aligned) and the disguised Flash-Lite judge per sentence stratum. Left of the dashed line: the four strata of development set E (CTRL $n=386$, L20 $n=821$, L25 $n=873$, EXC $n=606$). On short FOLIO controls consensus has no advantage (CTRL: $\Delta=-0.07$ [-0.23, 0.10]). The gap is significant on L20 (+0.11) and largest on exception sentences (EXC: +0.20). Right of the dashed line: R.COMP under the SIG condition (templated long rule sentences, $n=1,904$), where consensus reaches 0.954 versus the judge’s 0.587.

5.4 How Many Families Are Needed?

The consensus AUROC rises steeply from 1 to 3 families and plateaus thereafter. The optimal cross-fitted 3-family pool (DeepSeek-V3.1, Mistral-Small-3.2, GPT-4.1-Mini) achieves OOF AUROC 0.785, matching the full 9-family pool at 0.784. Leave-one-family-out analysis shows a minimum delta of -0.011 : no single family is indispensable.

Table 2: Consensus AUROC as a function of the number of peer families k .

Families (k)	AUROC	Cost/sentence
1	0.685	\$0.0003
3 (best pool)	0.785	\$0.0008
5	0.783	\$0.0012
7	0.783	\$0.0015
9 (full pool)	0.784	\$0.0019

In the long-sentence tercile, $k_{95} = 5$ families are needed for 95% of full-pool performance, reflecting greater vocabulary divergence.

5.5 Sensitivity to Error Types

On the PERTURB suite (4,234 typed mutations plus 868 controls, 300 bases), the within-base AUROC measures each metric’s ability to distinguish a known mutation from its correct base:

Table 3: Within-base AUROC on the PERTURB suite.

Metric	Within-base AUROC
PEER+TEXT fusion	0.948
Consensus (c_align)	0.866
FOL-Triage text layers	0.845
Round-trip NLI	0.749
Flash-Lite judge (disg.)	0.680
Self-consistency (SC-5)	0.666
Local Qwen3-8B judge	0.653

Consensus is polarity-symmetric: the absolute difference in recall between DOWN and UP perturbations is below 0.01 across all operators. Local judges and L3 miss DOWN edits (recall difference -0.12 to -0.15).

5.6 Negative Results

Pilot structural metrics. The user’s initial structural metrics (joint satisfiability conflicts, arity inconsistency, shape agreement, dangling predicates, undeclared predicates) achieve AUROCs between 0.500 and 0.533 on the evaluation screen. These metrics measure consistency and parseability, not faithfulness to the text.

Round-trip reformalisation. Re-translating the verbalized formula back to FOL and checking z3 equivalence with the original achieves AUROC 0.513, near chance, because 76% of items tie at the same score.

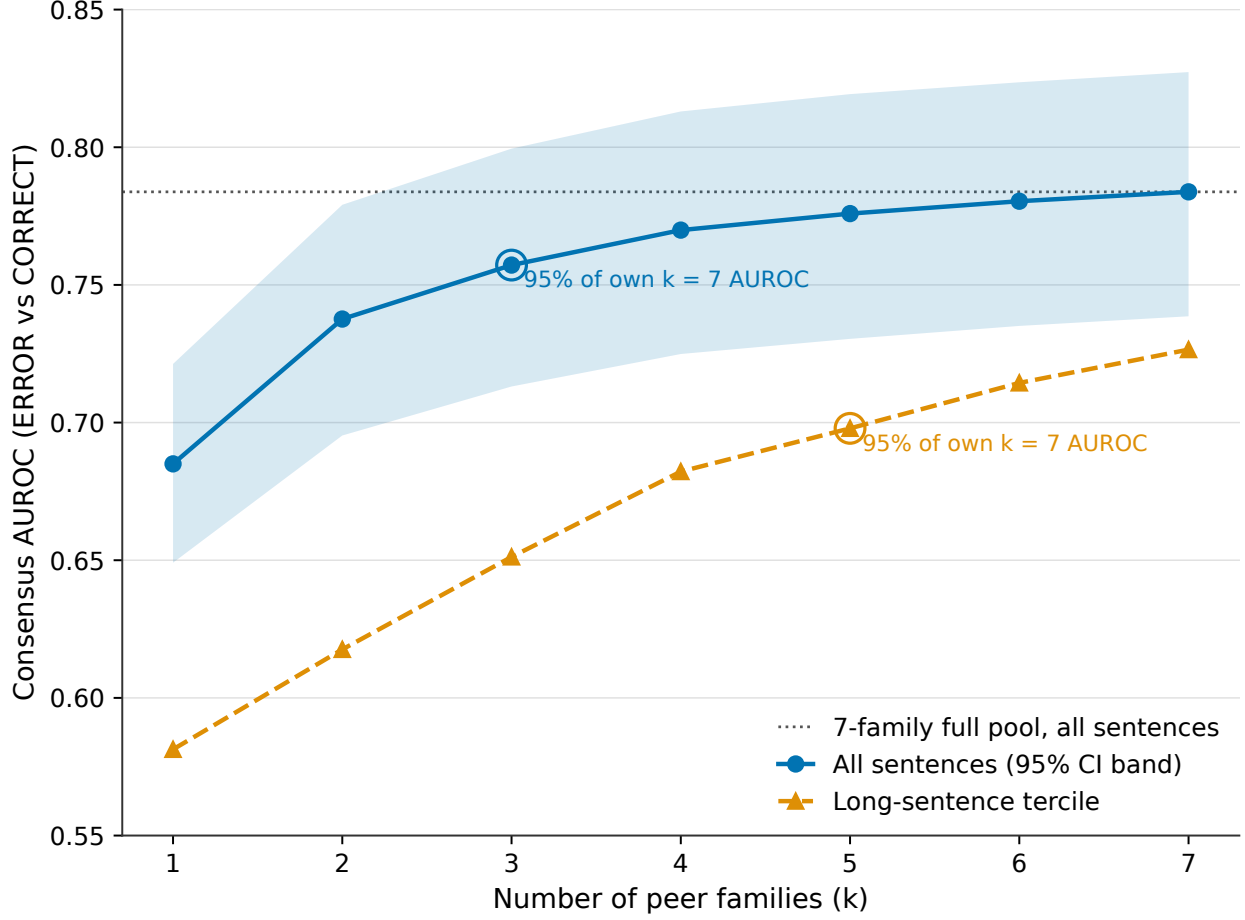


Figure 4: Consensus AUROC as a function of the number of peer families k on the development set ($n=2,014$ candidates from 186 sentences with all seven peer families available). Each point averages over 50 random k -subsets. Blue solid line: all sentences; orange dashed line: long-sentence tercile. Dotted grey line: full 7-family pool (AUROC 0.784). AUROC rises steeply from $k=1$ (0.685) to $k=3$ (0.757, 96.6% of full pool) and gains little thereafter. The long-sentence tercile starts lower and keeps climbing, needing $k=5$ families to reach 95% of its own full-pool value.

Candidate-signature consensus. Providing peers with the candidate’s own symbol list (to reduce vocabulary divergence) lowers correct-candidate divergence ($d = 0.139$ vs. 0.323 for free peers) but sharply raises error endorsement ($e = 0.459$ vs. 0.173): peers anchored on the candidate’s symbols reproduce its errors. Stratified AUROC drops to 0.614 $[0.49, 0.74]$, significantly below free-peer consensus at 0.774 . This confirms that consensus requires independent translations.

Fusion does not beat consensus alone. Fusing consensus with text-based FOL-Triage layers (PEER+TEXT) achieves stratified AUROC 0.753 on E, only $+0.011$ $[-0.012, 0.038]$ above consensus alone, not significant. The text layers add no discriminative power beyond what consensus already captures, once the stratum-composition confound is removed.

6 Discussion

6.1 Why Consensus Works

The consensus metric rests on a premise from N-version programming [Eckhardt and Lee, 1985, Knight and Leveson, 1986]: when independent implementations produce the same output, it is more likely to be correct. Errors scatter across the space of wrong formulas, while correct translations converge on the same logical meaning. On our data, the scatter is quantifiable: 76% of incorrect peer outputs for a given sentence are unique (a scatter index of 0.760), while correct translations cluster (scatter index 0.159), a ratio of 4.45 .

This scatter premise holds more strongly for longer sentences. The error endorsement rate e falls with the number of conditions, consistent with coincident-failure theory’s prediction that correlated failures decrease as input complexity rises [Eckhardt and Lee, 1985, Liu, 2026]. At the same time, correct-candidate divergence d rises with sentence length, because longer sentences elicit more diverse vocabulary choices. The net effect is that binary consensus (majority vote) degrades with length (the delta in $e + d$ between long and short terciles is $+0.356$ $[0.198, 0.493]$), but graded consensus, which uses the continuous fraction rather than a threshold, captures the additional information in partial agreement and maintains its advantage.

6.2 The Vocabulary Problem

The consensus metric’s central limitation is vocabulary dependence. Two formulas expressing the same meaning with different predicate names are not λ -equivalent, producing false alarms on correct candidates. Under synonym renaming, the vocabulary-aligned consensus false-alarm rate is 0.565 (on the PERTURB controls); under nonce renaming (arbitrary new predicate names), it reaches 0.765 .

We tested two approaches to mitigate this. A name-free matching algorithm that maps formula clauses by structural role (quantifier structure, argument count, nesting depth) without using predicate names achieves lower rename false-alarm rates (0.132 for nonce renames) but also lower overall AUROC (0.686 vs. 0.741 for vocabulary-aligned consensus). A gloss-gated hybrid that accepts name-free matches only when an LLM confirms that the renamed predicates preserve meaning failed its development gate (checker balanced accuracy 0.780 , below the 0.90 threshold). The fundamental tradeoff persists: loosening the equivalence criterion to tolerate vocabulary differences also accepts more errors.

6.3 Limitations of Confirmation

The pre-registered confirmation on a fresh held-out set (E2) was budget-stopped before reaching testability. Of 550 planned sentences, only 110 received panel labels; the primary long-sentence stratum collapsed to 94 rows (76 ERROR, 18 CORRECT), yielding a non-significant delta. On the 273 testable rows (dominated by ProverQA-derived sentences with prover-built gold), consensus achieves AUROC 0.890 but ties the local 8B judge (0.913, delta +0.003 [-0.042, 0.052]) and adds nothing over the S4 stack (-0.009 [-0.034, 0.014]). This stratum differs from the development set in sentence structure and gold provenance, so the non-result does not contradict the T1 confirmation but does bound the generalization claim. The confirmed advantage is specific to sentences with ≥ 20 words and ≥ 3 conditions.

7 Limitations

Vocabulary dependence. Consensus requires that correct translations share enough predicate vocabulary for the aligner to bridge. When model families adopt systematically different naming conventions, correct candidates are penalized. This problem is structural: any metric based on solver equivalence inherits it.

Error endorsement ceiling. When multiple families reproduce the same error, consensus cannot detect it. On our data, 38% of errors are endorsed by the majority (12.4% at the majority threshold with 9 families). Peer-endorsed errors are concentrated in structural error types (57%) rather than polarity errors (5%).

Label protocol sensitivity. The metric’s advantage over judges depends on the label protocol. Under panel-adjudicated intent labels, the frontier judge’s advantage over the cheap judge grows by +0.166 [0.086, 0.244], and the FOL-Triage text layers gain more than consensus. Singh et al. [2026] report that the LLM judge beats the trained verifier when labels switch from z3-reference to panel intent. Our primary labels are z3-equivalence to audited gold, which may overcount vocabulary mismatches as errors.

Short sentences. On the FOLIO CTRL stratum (short sentences, < 12 words, ≤ 1 condition), consensus shows no advantage over the judge (-0.069 [-0.23, 0.10]). The advantage concentrates in the long, conditioned, exception sentences that are the hardest to formalize.

Rename invariance unsolved. Every vocabulary bridge we tested either lowers discriminative power (name-free matching) or admits too many false errors (gloss-gated matching). A practical deployment would need to restrict the peer pool to a shared vocabulary, which reintroduces the ontology dependence the metric is designed to avoid.

Confirmation incomplete. Budget constraints prevented full confirmation on a fresh sample. The development-set results are the strongest evidence; the single testable confirmation cell shows parity rather than superiority.

8 Conclusion

Cross-family solver consensus is a viable gold-free metric for NL-to-FOL faithfulness, particularly for long, conditioned sentences where LLM judges degrade. Three peer families suffice, at a cost under \$0.002 per sentence. The metric’s vocabulary dependence is its principal limitation and remains unresolved. The meta-evaluation framework, including the 8,507-row development set, the typed perturbation suite, and the reusable consensus library, is released to support further work on autoformalization evaluation.

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